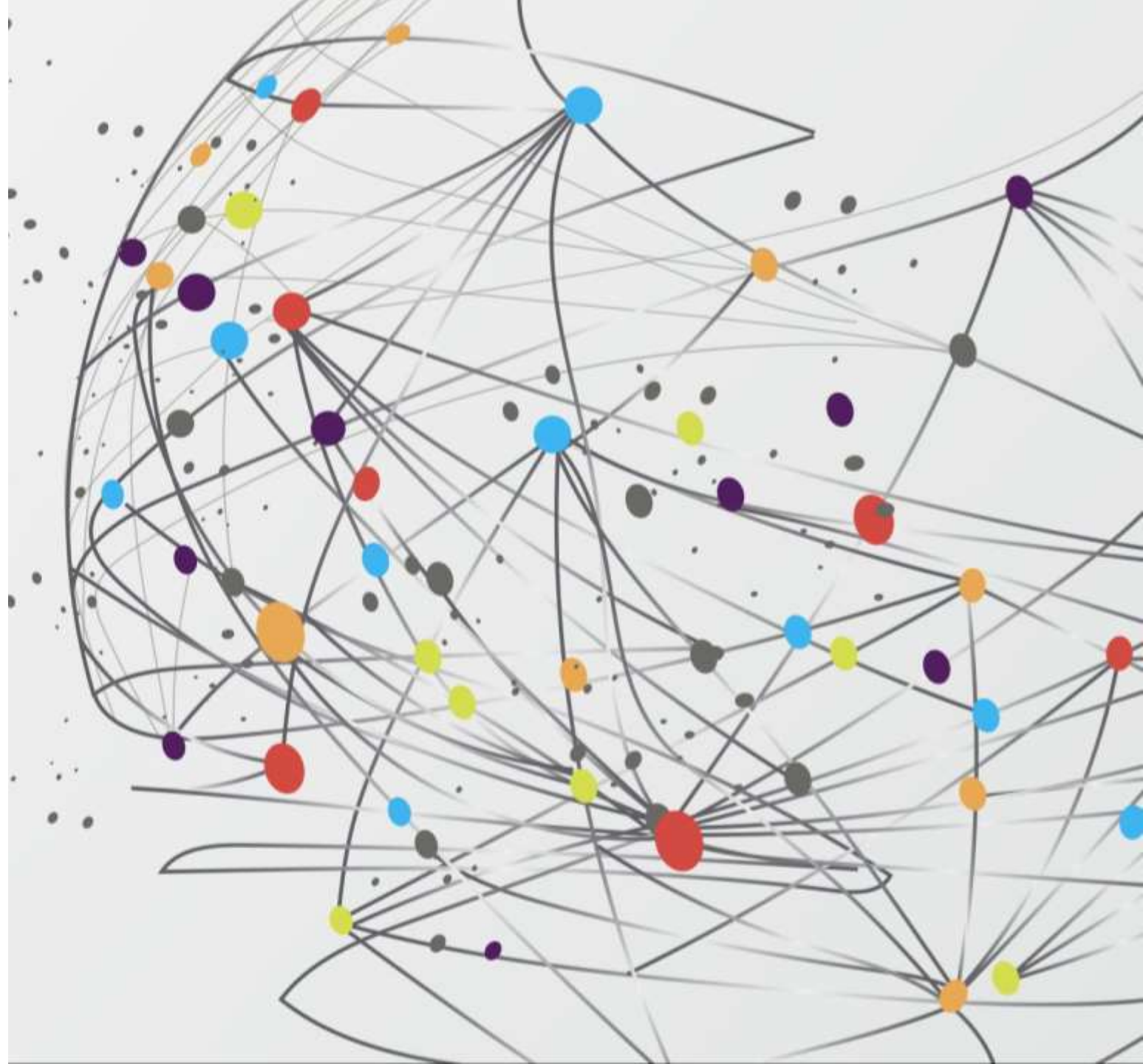

SİBER GÜVENLİĞE GİRİŞ

BÖLÜM 11 MOBILE DEVICE SECURITY

Dr. Öğr. Üyesi Ahmet Kaşif
ahmet.kasif@btu.edu.tr



11.1 INTRODUCTION

11.1.1 THE MOBILE REVOLUTION AND ITS SECURITY CHALLENGES

Mobile Usage Statistics: Billions of people worldwide use smartphones. Mobile online shopping has surpassed desktop shopping. Mobile website visits account for a majority in many regions.

Increased Threat Landscape: The popularity of mobile devices has led to a dramatic rise in security issues targeting these platforms.

11.1.2 THE GROWING MENACE OF MOBILE THREATS

McAfee Mobile Threat Report Q1 2018 Highlights:

- Significant increase in ad click fraud.
- Substantial rise in Trojan attacks, Botnet activities, spyware, and banking Trojans.

Financial Impact: The global market value of ad click fraud alone was estimated to be enormous.

- **Conclusion:** Mobile security is a critical and rapidly evolving area within cybersecurity.
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11.1.3 MAJOR THREATS TO MOBILE SECURITY

Major Mobile Threats:

- Increasing menace of ad click frauds
 - Increased Trojan attacks
 - BotNet activities
 - Crypto-mining malware attacks
 - Spyware attacks
 - Root malicious attacks
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Alarming Malware Growth: Millions of new mobile malware are detected annually, with the numbers increasing rapidly.

- **Future of Cybersecurity:** Mobile security is poised to be a central focus of cybersecurity efforts in the coming years.

11.2 IMPORTANCE OF MOBILE SECURITY

11.2.1 THE CENTRAL ROLE OF MOBILE DEVICES IN OUR LIVES

Widespread Usage: Billions of people worldwide own smartphones.

- **Multifunctional Nature:** Smartphones are used for a vast array of daily activities.

11.2.2 WHY SECURING YOUR MOBILE DEVICE IS ESSENTIAL

Reasons for Mobile Security:

- **Capturing Memories:** Used as a camera for photos and videos of personal moments.
 - **Communication Hub:** Facilitates voice calls, SMS, chats, video calls, and emails.
 - **Portable Computer:** Enables internet browsing and online shopping.
 - **Navigation Tool:** Provides guidance through maps and GPS.
 - **Entertainment Center:** Serves as a platform for various forms of entertainment.
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- **Digital Wallet:** Used for making payments in physical stores.
 - **Access Management:** Controls physical and digital access.
 - **Storing Personal Information:** Holds valuable contacts and personal data.
 - **Temporary Data Storage:** Used for storing various types of temporary data.

Consequences of Security Breaches: A compromised mobile device can lead to breaches of privacy, data loss, and other significant issues.

- **Call to Action:** Ensure your smartphone is fully protected from cybersecurity threats.
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11.3 TYPES OF MOBILE PLATFORMS

11.3.1 THE EVOLUTION OF MOBILE OPERATING SYSTEMS

Historical Categories:

- **(1973–1993): Embedded OSs:** Basic systems for function control in early mobile phones.
- **(1993–1999): Independent Mobile OSs:** Introduction of early smartphone OSs like Newton OS, Palm OS, Symbian OS.
- **(2000–2010): Rise of Smartphones:** Key OS launches including Windows CE, Blackberry, iOS, Android, webOS.
- **(2010–Present): Platform Competition:** Dominated by Android and iOS, with ongoing competition and newer versions.

Current Market Leaders: Android and iOS are the dominant players in the mobile OS market today.

11.3.2 THE DOMINANCE OF ANDROID AND IOS

Global Market Share (January 2019):

- **Android:** 74.45% (Market Leader)
- **iOS:** 22.85%
- **Other OSs (KaiOS, Windows, Samsung OS, etc.):** Smaller market shares.

Focus: This chapter will primarily focus on Android and iOS due to their market dominance.

11.3.3 ANDROID OPERATING SYSTEM

Development: Based on a modified Linux kernel and primarily developed using Java, C, and C++.

Launch: The first Android version was released in September 2008.

Latest Version (as of text): Android 9 (Pie), released in August 2018.

Google Services: Associated with proprietary services like Google Play, Chrome, Gmail, and Google Search.

User Base and App Ecosystem: Over 2 billion monthly active users and millions of apps on the Google Play Store.

Major Manufacturers: Samsung, Huawei, Xiaomi, Motorola, Oppo, LG are key Android phone manufacturers.

11.3.4 iOS OPERATING SYSTEM

Development: A proprietary system written in Objective-C, C++, C, and Swift.

Launch: The first version was released on June 29, 2007.

Latest Stable Version (as of text): iOS 12.1.4 (February 2019).

OS Layers: Based on a multi-layered architecture including:

- *Core operating system layer

- *Core Services layer

- *Media layer

- * Cocoa Touch layer

App Ecosystem: Over 2 million apps available on the Apple App Store, the second largest marketplace.

Target Market: Popular among higher-income users, with a significant number of paid applications.

11.4 LOCKED BRANDED PHONES

11.4.1 UNDERSTANDING LOCKED BRANDED PHONES

Branding and Locking: Mobile phone companies sometimes modify the firmware of their devices to restrict usage to a particular carrier's services.

Business Model: These phones are often offered at a lower initial cost but are locked to a specific operator for a set period, binding the user to that provider.

SIM Restriction: Typically feature a single SIM slot and do not allow the use of SIM cards from other carriers.

Pre-installed Applications: Often include built-in applications specific to the service provider.

11.4.2 SECURITY CONCERNS WITH LOCKED BRANDED PHONES

Data Routing: User data is primarily routed through the single, contracted carrier.

Remote Access: In some instances, locked phones may have remote access enabled by the service provider.

Increased Security Concerns: The combination of single-carrier data routing and potential remote access can increase security vulnerabilities.

Recommendation: Purchasing unbranded phones is generally recommended to maintain greater control over your data and security.

Benefits of Unbranded Phones:

- Your data and personal information are not tied to a single carrier.
- You have the freedom to choose and use any security applications you prefer.

11.5 ANDROID PHONE SECURITY GUIDELINES

11.5.1 ANDROID SECURITY: LEVERAGING A ROBUST FOUNDATION

Linux Kernel Strength: The underlying Linux kernel is known for its power and security.

Open Source Benefits: A large community of developers continuously contributes to improving Android's security by identifying and patching vulnerabilities.

Google's Role: Google's development teams actively provide quick fixes for Android OS vulnerabilities.

The Update Challenge: A significant issue is the delay in update rollouts by phone manufacturers.

11.5.2 ESSENTIAL SECURITY GUIDELINES FOR ANDROID USERS

Key Security Practices:

- **Latest OS Version:** Purchase phones running the newest Android version.
 - **Encrypted Storage:** Enable data encryption for internal and external memory.
 - **Account Manager:** Save credentials in secure Account Manager tools.
 - **Trusted App Sources:** Download apps ONLY from the Google Play Store.
 - **Antivirus/Anti-Spyware:** Install and use paid versions of reputable security software.
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- App Permissions:** Carefully review and grant only necessary app permissions.
 - Privacy Policies:** Read the privacy policies of the apps you use.
 - Regular App Updates:** Keep all your Android applications updated.
 - Information Exposure:** Ensure apps aren't unnecessarily sharing your data.
 - Avoid Public Wi-Fi:** Refrain from using insecure public wireless networks.
 - READ_LOGS Permission:** Deny apps the READ_LOGS permission.
 - VPN Usage:** Use Virtual Private Networks (VPNs) for secure browsing.
 - Vendor Update Speed:** Research how quickly manufacturers release OS patches before purchasing.
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- Strong Screen Lock:** Use strong passwords (biometrics preferred over patterns/PINs).
 - Two-Factor Authentication:** Enable 2FA for your Google account and linked services.
 - Separate 2FA Numbers:** Use a different phone number for 2FA than the one in your device.
 - Device Encryption:** Enable Android's built-in device encryption.
 - Password Management Tools:** Utilize password manager apps for secure credential storage.
 - Disable Unused Wireless:** Turn off Wi-Fi and Bluetooth when not in use.
 - Uninstall Unused Apps:** Remove applications you no longer need.
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- Avoid SMS for Sensitive Info:** SMS messages are not encrypted on carrier networks.
 - Secure Communication Apps:** Use encrypted communication apps (e.g., WhatsApp).
 - Device Security:** Never hand over your phone to unknown individuals.
 - Ad Blocker:** Install an ad blocker to prevent malicious ad-related activities.
 - Pop-up Blocker:** Enable pop-up blockers in your browser.
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- Disable JavaScript (Carefully):** Consider disabling JavaScript in your browser for increased security (may affect website functionality).
 - Remote Device Locator:** Enable "Find My Device" features.
 - Automatic Backup:** Activate cloud backups for data recovery.
 - Factory Reset Before Sale:** Perform a factory reset before selling your phone.
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11.5.3 ENABLING BIOMETRIC SECURITY ON ANDROID

General Steps:

- 1. Swipe Down for Quick Settings:** Swipe from the top of your screen.
 - 2. Tap Settings Icon:** Look for and tap the gear (⚙️) icon.
 - 3. Navigate to Security:** Find and tap options like "**Lock Screen and Security**" or "**Security**".
 - 4. Select Screen Lock Type:** Tap on "**Screen Lock Type**".
 - 5. Choose Fingerprints:** Scroll down and select "**Fingerprints**".
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6.Set Up Fingerprint Security: Follow the on-screen wizard.

7.Alternative Lock (Password/PIN/Pattern): You'll likely be prompted to set up a backup lock method before enabling fingerprints. Choose your preferred option (e.g., "**Password**").

8.Add Backup Lock: Enter and confirm your backup password/PIN/pattern.

9.Access Fingerprint Scanner: Return to the "**Lock Screen and Security**" or "**Security**" page and tap "**Fingerprint Scanner**".

10.Scan Your Finger: Place your finger on the designated sensor (often the home button or back of the phone) as guided until the scan is 100% complete.

11.Fingerprint Added: Your fingerprint lock is now enabled along with your chosen backup security method.

11.6 IPHONE SECURITY GUIDELINES

11.6.1 IPHONE SECURITY: A STRONG ECOSYSTEM

Apple's Commitment: A dedicated team ensures quick responses to security vulnerabilities.

Proactive User Measures: Despite iOS's strength, users should follow security guidelines to prevent breaches.

11.6.2 ESSENTIAL SECURITY GUIDELINES FOR IPHONE USERS

Key Security Practices:

- **Siri Security:** Be mindful of leaving your unlocked phone unattended to prevent unauthorized Siri access.
 - **iCU for Enterprise:** Utilize iPhone Configuration Utility in corporate environments for managed security.
 - **Strong Passwords:** Create complex and unique passwords for all accounts.
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- Password Manager:** Use a reputable password manager for enhanced security.
 - Authorized Apps Only:** Avoid installing unauthorized or jailbroken applications.
 - Disable On-Screen Notifications (Sensitive Info):** Prevent sensitive information from appearing on the lock screen.
 - Two-Factor Authentication:** Enable 2FA for your Apple ID and iCloud.
 - Disable iCloud Auto-Sync (If Concerned):** Manage automatic data synchronization carefully.
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- Disable Siri on Locked Screen (If Concerned):** Prevent Siri access when the phone is locked.
 - Use VPN:** Employ a Virtual Private Network for secure internet browsing.
 - Disable Browser Auto-Fill:** Prevent automatic form filling in web browsers.
 - Limit App Access to Personal Info:** Control which apps can access your contacts, location, etc.
 - Disable Browser Cookies (Privacy Focus):** Manage or disable browser cookies.
 - Disable Auto Wi-Fi Connection:** Prevent automatic connection to unknown Wi-Fi networks.
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- **Use Secure Networks:** Avoid public and untrusted Wi-Fi networks.
 - **Turn Off Unused Protocols:** Disable features like Bluetooth or Airdrop when not in use.
 - **Encrypt Local Files (If Applicable):** Utilize any available file encryption options.
 - **Backup to iCloud:** Regularly back up your data to iCloud.
 - **Use Encrypted Communication:** Prefer apps with end-to-end encryption (e.g., Signal, WhatsApp).
 - **Update Apps Regularly:** Keep all your iPhone applications updated.
 - **Keep iOS Updated:** Install the latest iOS software updates promptly.
 - **Use Common Sense:** Exercise caution while browsing and shopping online.
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11.6.3 ENABLING BIOMETRIC SECURITY: TOUCH ID

Initial Setup: Often activated during the iOS Setup Assistant.

Manual Configuration Steps:

1. Open **Settings** and select **Touch ID & Passcode**.
 2. Tap **Turn On Passcode** and enter a unique four-digit passcode.
 3. Go back to **Touch ID & Passcode** and tap **Add a Fingerprint....**
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4. Follow the on-screen instructions to place your finger on the Home button in various positions for scanning.

5. The progress will be indicated on the screen.

6. Once the initial scan is complete, you may be prompted to **Adjust Your Grip** and repeat the scanning process with different hand positions.

7. A **Success** screen will appear when the fingerprint is successfully recorded.

8. You can now use your fingerprint to unlock your iPhone. Press the lock button to sleep the screen and then the Home button with your registered finger to unlock.

Note: You can register multiple fingerprints

11.7 WINDOWS PHONE SECURITY GUIDELINES

11.7.1 WINDOWS PHONE SECURITY: FAMILIAR AND INTUITIVE

Familiarity: Many users are already acquainted with Windows OS security features and guidelines from their computer experience.

Focus: Applying these known security principles to the mobile environment is key.

11.7.2 KEY SECURITY GUIDELINES FOR WINDOWS PHONE USERS

Key Security Practices:

- **Latest OS Version:** Purchase and use the newest Windows Phone OS version.
 - **Regular OS Updates:** Keep your OS up to date with all new updates and releases.
 - **Trusted App Store:** Download mobile applications exclusively from the Microsoft Store.
 - **Paid Apps (Optional):** Consider paid apps to avoid advertisements and potential pop-up annoyances.
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- Keep Apps Updated:** Ensure all your installed Windows apps are regularly updated.
 - Install Security Software:** Use a reputable antivirus and anti-malware tool.
 - Multi-Step Authentication:** Activate multi-factor authentication for your accounts.
 - Screen Lock PIN:** Always use a strong screen lock PIN to secure your phone.
 - "Find My Phone":** Enable the "Find My Phone" feature for device location in case of loss.
 - Secure Browser Settings:** Use the Internet Explorer browser with strong security settings.
 - Disable Browser Cookies (Privacy):** Prevent your browser from storing cookies.
 - Avoid Public Networks:** Never use insecure public Wi-Fi networks.
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- Secure Websites:** Always browse and transact on secure (HTTPS) websites.
 - VPN Connection:** Use a VPN for secure browsing and online shopping.
 - Regular Data Backup:** Back up your important data regularly.
 - Hardware Security:** Utilize newer phones with security microchips or Trusted Platform Modules (TPMs).
 - Biometric Unlock:** Employ facial recognition, fingerprints, or iris scanning if available.
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- Enlightened Apps (Enterprise):** Use secure, managed applications in corporate environments.
 - Limit Remote Access:** Use remote access features only when absolutely necessary.
 - Trusted Shopping Sites:** Only make purchases from reputable online retailers.
 - Careful App Permissions:** Review and grant third-party application permissions cautiously.
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11.8 MOBILE APPLICATION MANAGEMENT

11.8.1 THE EXPANDING UNIVERSE OF MOBILE APPS AND THEIR RISKS

App Proliferation: Millions of apps exist, with hundreds of thousands added annually.

Third-Party Development: Apps are typically created by companies with their own business objectives.

Inherent Risks: These objectives can sometimes conflict with user security if guidelines aren't followed.

Exploitable Vulnerabilities: Hackers target weaknesses in app code and operations.

User Responsibility: Both individuals and enterprises must ensure apps are used securely.

11.8.2 THE ENTERPRISE CHALLENGE: BYOD AND MAM

BYOD Popularity: Enterprises are increasingly adopting BYOD to save costs.

BYOD Security Risks: Employee-owned devices with unmanaged apps can compromise enterprise data.

MAM Solution: Mobile Application Management software controls and secures applications and the enterprise data they access.

11.8.3 CORE ASPECTS OF MOBILE APPLICATION MANAGEMENT (MAM)

MAM Functions:

- Assessing app security measures.
 - Auditing app vulnerabilities.
 - Establishing controls over apps.
 - Managing app updates.
 - Checking for security policy violations.
 - Containerization of apps to isolate data.
 - App addition, modification, and deletion.
 - Blacklisting and whitelisting of apps and activities.
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11.8.4 GUIDELINES FOR PERSONAL MOBILE APPLICATION MANAGEMENT

Key Guidelines for Personal App Management:

- **Trusted Stores:** Download apps only from Apple App Store and Google Play.
 - **App Details and Reviews:** Read descriptions and user reviews before downloading.
 - **Permission Awareness:** Pay close attention to requested app permissions.
 - **Careful Installation:** Read every step of the installation wizard; avoid blindly clicking "Next."
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- Custom Installation:** Choose installation options that suit your needs.
 - Selective App Choice:** Be discerning when selecting mobile apps.
 - Deny Undue Permissions:** Do not install apps requesting unreasonable permissions.
 - Uninstall Unused Apps:** Remove applications you no longer use.
 - Regular Updates:** Update all applications promptly.
 - Enable Auto-Updates:** Configure apps to update automatically.
 - Install Security Software:** Use a reputable antivirus and anti-malware app.
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- Organize Apps:** Arrange apps for easy tracking.
 - Frequency-Based Arrangement:** Organize apps by how often you use them.
 - Containerization (Personal Use):** Explore apps that offer data isolation.
 - Encrypted Connections:** Use apps that communicate with servers over secure connections.
 - Automatic Key Manager:** Utilize password management tools for app logins.
 - Strong App Passwords:** Use strong, unique passwords for individual apps.
 - Report Abnormal Behavior:** Immediately remove any app exhibiting suspicious activity.
 - Share Your Feedback:** Leave reviews to help other users.
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